Venkata Lakshmi Parimala Pasupuleti

Forward Deployed Engineer · Python / LangGraph / LLMs · Agentforce & Agentic AI · Enterprise Consulting Delivery · Eval Infrastructure

venkatalakshmiparimalap@gmail.com | +1-703-261-5410 | Virginia, USA | LinkedIn | AgentProbe (GitHub) | GitHub

SUMMARY

Forward Deployed Engineer with 9 years of production software ownership. At Magizhs Tech Solution, designed and delivered a production LangGraph agentic AI system for an external client -- owned architecture, LLM selection, RAG layer, prompt engineering, output validation, and integration with enterprise data sources, end to end. Built AgentProbe -- an open-source eval platform used to validate AI agent outputs before production: CI-integrated regression gate, LLM-as-judge scoring, hallucination detection, agent tracing, and multi-provider LLM routing. Built LeaseI -- an 8-agent LangGraph system with RAG, structured data extraction, and HITL guardrails on regulated decisions, evaluated across 1,240 test cases at 93.6% precision. Seven years owning data pipelines, APIs, and enterprise compliance systems at Entrata (Python, JavaScript/TypeScript, PostgreSQL, RabbitMQ, AWS, 100K+ MAU). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, 2026).

SKILLS

AI & Agentic Systems: Python, LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, Large Language Models (LLMs), Generative AI, Prompt Engineering, Tool Calling, RAG Pipelines, Vector Databases, ChromaDB, Embeddings, HITL, Guardrails, Output Validation, Hallucination Detection, LLM Evaluation, LLM-as-Judge, Agent Tracing, Eval Infrastructure, LLMOps, Fine-tuning (HuggingFace, PyTorch), NLP, BERT, spaCy

Full-Stack & Data Pipelines: Python, JavaScript, TypeScript, Java, SQL, React.js, Angular, FastAPI, Flask, REST APIs, Microservices, Data Pipelines, ETL, Data Modeling, RabbitMQ, Redis, PostgreSQL, MySQL, Oracle, AWS (S3, EC2, RDS, Glue, Lambda), Azure, GCP, Docker, Kubernetes (K8s), CI/CD (GitHub Actions), Git, Linux, Agile/Scrum

Enterprise & Compliance: HIPAA Compliance, PHI Data Handling, RBAC, Libsodium, Distributed Systems, Observability, Performance Engineering, Code Review, Technical Mentorship

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI Engineer (Customer-Facing Delivery)

- Designed and delivered a production LangGraph agentic AI system for an external client -- owned the full technical roadmap from architecture through deployment: LLM selection, RAG layer over a regulatory corpus, prompt engineering strategy, output validation, and integration with enterprise data sources.
- Translated customer business requirements into a working agentic solution; shipped an eval harness alongside the product so every output was validated before reaching the client's production environment.
- Built validation guardrails enforcing compliance rules deterministically -- non-compliant outputs blocked before delivery, with grounded cited reasons for every rejection.

Entrata (PropTech SaaS · 100K+ MAU · enterprise data pipelines · HIPAA/PHI)
Sep 2018 - Aug 2024

Senior Software Engineer

- Owned compliance-critical data pipelines for 7 years (180,000 HUD records powering housing eligibility, rent, and tax-credit calculations for 100K+ units) -- zero violations; validation, reconciliation, and rollback built into every release cycle.
- Shipped 20+ full-stack production modules (Python, JavaScript/TypeScript, React, Angular, PostgreSQL, RabbitMQ) across 120+ Agile sprints; led Docker microservices redesign (50% scalability improvement, 30% deployment time reduction).
- Led AWS S3 document migration -- Libsodium-encrypted pipelines, 100% PHI compliance, zero security incidents; built Python/Bash automation eliminating 80% of manual operational workload.
- Optimized 200+ SQL queries (30% performance gain) through execution plan analysis, indexing, and stored procedure refactoring; mentored 5+ engineers; conducted 300+ peer code reviews.

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Fine-tuned a custom classification model architecture using HuggingFace Transformers and PyTorch; deployed inference API on Quart; presented model design and evaluation results to faculty.
- Built multi-source ML prediction models (Python, R, $R^2 = 0.977$); applied Isolation Forest anomaly detection across 3 research pipelines; delivered Python + database workshops to 600+ students.

PROJECTS

AgentProbe: AI Eval Infrastructure & LLMOps Platform
2025 | github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · LangChain · ChromaDB · CI/CD (GitHub Actions) · Multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama)

- Open-source eval platform: CI-integrated regression gate, LLM-as-judge scoring, hallucination detection, pass@k harness, agent tracing, and multi-provider LLM routing. Ships quality code and evaluates AI outputs with engineering rigor -- the infrastructure for knowing whether an agentic system is ready for production before it touches real users.

Leasel: 8-Agent Agentic System with RAG, HITL & Data Pipelines

2026 | github.com/venkatalakshmiparimala/Lease_complaince_rag

Python · LangGraph (8-agent) · FastAPI · RAG · OCR · ChromaDB · SQLite · Streamlit

- Production agentic AI system: 8 LangGraph agents with RAG retrieval, structured data extraction, prompt engineering, HITL guardrails on regulated decisions, and conditional routing. 93.6% precision, 87.9% F1 across 1,240 test cases. Architected data pipelines for OCR ingestion, section-aware chunking, and retrieval -- from sketch to deployed and observable in production.

AI on FHIR: HIPAA-Compliant Healthcare AI System

2024 | github.com/venkatalakshmiparimala/AI_on_FHIR

Flask · React · LLMs · FHIR APIs · spaCy · BERT · Docker · HIPAA · RBAC

- HIPAA-compliant AI system over structured PHI -- RBAC, NLP-to-FHIR API translation (EN/ES), LLM + BERT + embeddings integration; reduced clinical retrieval time ~60%.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — Graduated May 2026 | GPA: 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017